

Physics-Guided Visual Air-Quality Estimation

The complete project, explained from the ground up

A no-prior-knowledge walkthrough: every concept, every equation, every number, and how each piece builds on the last — enough to present and defend to a professor.

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Appendix A: the number sheet · Appendix B: the equation sheet. Nine figures and the key equations are typeset throughout.

Physics-Guided Visual Air-Quality Estimation

The complete project, explained from the ground up

A sequential, no-prior-knowledge-assumed walkthrough of the entire project — every concept, every equation, every number, and how each piece builds on the last. Written so any team member can present and defend the whole thing to a professor without a single gap. Read it top to bottom; each section is set up by the one before it.

Team: Keyaan · Anup · Lamiya · Asif. Generated 2026-09-28.

How to use this document

- **Read in order the first time.** The story is a chain: metrics set up leakage, leakage sets up the

honest model, the model sets up calibration, and so on. Skipping ahead breaks the logic.

- **Three passes.** Each concept is explained (1) with an everyday intuition, (2) with a precise

definition/equation, and (3) with *how this project uses it and the exact numbers*. If you only have a few minutes, read the **bold sentences** and the **grey callout boxes**.

- **Boxes.** A box beginning "**Worked example**" walks through real arithmetic. A box beginning

"**Key idea**" is the one sentence to memorise. A box beginning "**Trap**" is a mistake to avoid (and a likely professor probe).

- **The defense-ready summary** is Section 12 (five-minute story + ~50-question Q&A + traps). The exact

numbers are collected in the Appendix.

1. Orientation — what we built, and why it's unusual

1.1 The one-sentence thesis

We estimate air quality (a US-EPA **AQI** value) from a single street photo; we do it **honestly** (a calibrated range, a refuse-to-answer option, and a physical error ceiling); and along the way we **prove** that a popular benchmark's high accuracy is inflated by data leakage.**

Most "predict X from a photo" projects chase one headline accuracy number and stop. Ours is deliberately different: **the contribution is honesty and measurement**, not a bigger number. That is the single most important framing to have ready, because the natural first question — **"but your accuracy is lower than the benchmark, so isn't your project worse?"** — is answered by this framing (Section 4).

1.2 The four honest behaviours (= the four contributions)

- **C1 — a calibrated range, not a single number.** Instead of "AQI = 137", we output a low–median–high

interval with a *mathematical guarantee* that the truth falls inside about 90% of the time.

- **C2 — an error ceiling.** We estimate the best R^2 that *any* model could possibly achieve on this

data, so we know how much of the remaining error is genuinely unbeatable versus fixable.

- **C3 — abstention.** The system **refuses to answer** on photos it should not guess on (night or

featureless; indoor is a documented limitation of the deployed station-invariant gate), and we measure how well it refuses.

- **Leakage-safe evaluation.** We evaluate in a way that cannot cheat, and we *measure* how much a common

shortcut inflates the score — the project's headline scientific finding.

1.3 The units rule (memorise this — it trips everyone up)

The label we predict is a **US-EPA Air Quality Index (AQI)** value: a unitless index that runs roughly **1–530**. It is **not** a concentration in micrograms per cubic metre ($\mu\text{g}/\text{m}^3$). Every error number in this document — MAE, RMSE, interval width — is in **AQI points**. We touch $\mu\text{g}/\text{m}^3$ only inside the error-ceiling analysis (C2), where we convert it to AQI with the official EPA table.

Key idea. The output is an *index* (AQI 1–530), not a raw concentration. All errors are in AQI points. Say this if a professor asks about units.

1.4 The fact that drives half the project

Each photo's label is the **daily-average** AQI for that location, but a photo is a single **instant**. Pollution rises and falls through a day, so the label is a slightly "smudged" answer key: even a perfect photo-reader would disagree with the daily average by the day's within-day swing. This one fact is the root of the **error ceiling** (Section 8) — hold onto it.

Worked example (why the smudged key matters). Suppose a street's pollution over a day is [80, 120, 160, 120, 80] AQI at five moments; the **daily average** (the label) is 112. A photo taken at the 160 moment shows "160", but it is scored against 112 — a 48-point "error" that is *not* the model's fault*. Averaging away time creates unavoidable error.

2. Machine learning from absolute zero

Everything later stands on these foundations. If you are shaky on any one of them, a professor's first question can unravel the whole defense — so we build them slowly and completely.

2.1 What a "model" is

A **model** is a mathematical function with adjustable numbers inside it, called **parameters** (or **weights**). You feed it an input and it computes an output. The simplest example is a straight line **output = $w \cdot \text{input} + b$** , with two parameters **w** (slope) and **b** (intercept). A deep network is the same idea with *millions* of parameters arranged in layers, capable of representing very complicated functions.

- **Feature:** a piece of the input the model reads. Here: the pixels of a photo, plus two physics maps

we compute (Section 5.1).

- **Label (target):** the correct answer for a training example. Here: the photo's AQL.
- **Parameters/weights:** the adjustable numbers the model learns.

2.2 How training works (gradient descent, intuitively)

- We define a **loss**: a single number measuring how wrong the model's outputs are on a batch of

training examples (small loss = good).

- Training repeatedly (a) runs the model, (b) measures the loss, (c) computes how the loss would change

if we nudged each parameter a tiny bit (the **gradient**), and (d) nudges every parameter a small step in the direction that *reduces* the loss. Repeat millions of times. This is **gradient descent**; the algorithm that does the nudging is the **optimiser** (we use one called AdamW).

- **Epoch:** one full pass over the training data. We train for up to 60 epochs.

Key idea. Training = automatically tuning millions of parameters to make the loss small, one small gradient step at a time.

2.3 Generalisation, overfitting, and the three data splits

- **Generalisation** = performing well on inputs the model **never saw during training**. This is the

entire point; a model that only does well on its training data is useless.

- **Overfitting** = memorising the training examples (including their noise) instead of learning the

underlying pattern. Training looks great, new data fails.

- To detect and control this we split data into **train** (fit the parameters), **calibration/validation**

(tune choices and, here, build the interval guarantee), and **test** (evaluated once, at the end, for the honest score). **Leakage** (Section 3) is a sneaky way apparent success on "test" is really memorisation — the central villain of this project.

2.4 Regression vs classification

- **Classification** picks a category (cat vs dog; or one of the six EPA AQI bands).
- **Regression** predicts a number. Predicting AQI is **regression**. (We also report an EPA-category

accuracy as a secondary, classification-style view, by bucketing predicted AQI into the six EPA bands.)

2.5 Neural networks and CNNs

- A **neuron** computes a weighted sum of its inputs, adds a bias, and passes it through a simple

non-linear function (an **activation**, e.g. ReLU which is $\max(0, x)$). Non-linearity is what lets stacked layers represent complex patterns.

- A **neural network** stacks many neurons in **layers**; "**deep**" just means many layers.
- A **convolutional neural network (CNN)** is the variant built for images. Instead of connecting every

pixel to every neuron, it slides small learnable **filters** (e.g. 3×3) across the image. Early filters learn edges; deeper layers combine them into textures, then object parts, then whole-scene properties (like *haze*). **Pooling** layers shrink the spatial size so the network sees larger context. The result of the CNN body is a compact **feature vector** summarising the image.

A CNN turns a photo into a feature vector; heads make the prediction

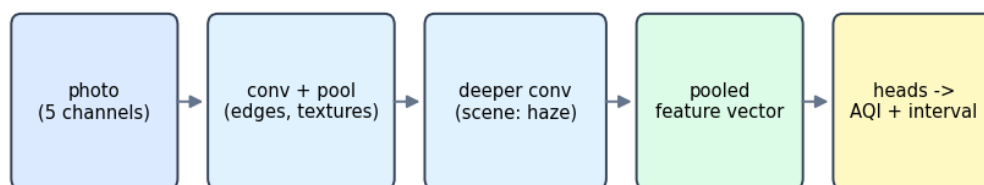


Figure 1. How a CNN reads an image (Section 2.5).

2.6 Backbones, pretraining, and transfer learning

- A **backbone** is a CNN body that turns an image into a feature vector. We use **EfficientNet-B0**

(~5.3 million parameters) — small, efficient, and the strongest backbone the source paper found on this dataset.

- **Pretrained** means the backbone was already trained on a giant generic dataset (**ImageNet**,

~1.2 million labelled photos), so it already "knows" edges and textures.

- **Transfer learning / fine-tuning** = we start from those pretrained weights and continue training on

our task. This works far better than random initialisation because our dataset (~7k training photos) is too small to learn good visual features from scratch. The backbone brings general vision knowledge; we adapt it to read haze.

Trap. "Why not a bigger network?" → The source paper found EfficientNet-B0 *beats* ResNet50 and ViT on this ~7k-image dataset; with little data, extra capacity overfits. The bottleneck here is data and label noise, not model size.

2.7 The measuring sticks (know these cold — the whole story is told in them)

All in AQI points. Notation: **y** = truth, **yhat** = prediction, **n** = number of examples, **ybar** = mean of the truths.

MAE (Mean Absolute Error) — the average size of the miss:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

"On a typical photo we're off by this many AQI points." MAE does **not** depend on how spread out the test set is, which makes it the most honest single accuracy number.

RMSE (Root Mean Squared Error) — like MAE but squares the errors first, so large misses are punished more:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$\text{RMSE} \geq \text{MAE}$ always. If $\text{RMSE} \gg \text{MAE}$, a few big errors dominate.

Worked example (MAE vs RMSE). Errors on five photos: [5, 5, 5, 5, 60]. $\text{MAE} = (5+5+5+5+60)/5 = 16$. $\text{RMSE} = \sqrt{((25+25+25+25+3600)/5)} = \sqrt{(3700/5)} = \sqrt{740} = 27.2$. The single 60-point miss barely moves MAE but nearly doubles RMSE — that is RMSE "punishing" the big miss. Our honest split has MAE 54 and RMSE 95, a large gap, which tells us a subset of photos (the extreme-AQI tail) is missed badly.

R² (the coefficient of determination) — the headline score. It answers: "what fraction of the variation in the truth does the model explain, compared to the dumbest baseline of always guessing the average?"

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}}$$

- **SS_{res}** = the model's squared error. **SS_{tot}** = the squared error of always guessing the mean **ybar**.
- **R² = 1** perfect; **R² = 0** exactly as good as guessing the mean; **R² < 0** worse than guessing

the mean (yes, it can be negative — not a bug).

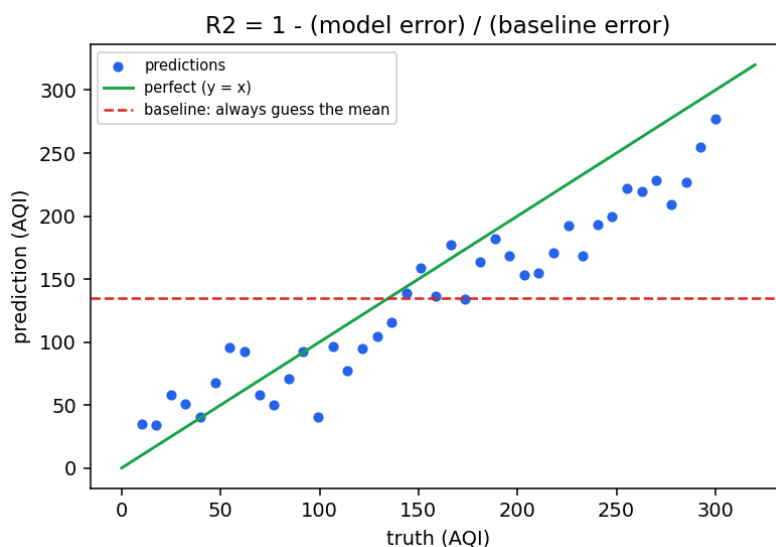


Figure 2. What R² measures (Section 2.7).

Worked example (R^2). Truths [10, 20, 30, 40, 50], mean $\bar{y} = 30$, so

$SS_{\text{tot}} = 20^2 + 10^2 + 0 + 10^2 + 20^2 = 1000$. Model A predicts perfectly $\rightarrow SS_{\text{res}} = 0 \rightarrow R^2 = 1$. Model B always says 30 $\rightarrow SS_{\text{res}} = 1000 \rightarrow R^2 = 0$. Model C predicts [14, 18, 30, 42, 46] $\rightarrow SS_{\text{res}} = 4^2 + 2^2 + 0 + 2^2 + 4^2 = 40 \rightarrow R^2 = 1 - 40/1000 = \mathbf{0.96}$. Model D (inverted) [50, 40, 30, 20, 10] $\rightarrow SS_{\text{res}} = 40^2 + 20^2 + 0 + 20^2 + 40^2 = 4000 \rightarrow R^2 = 1 - 4000/1000 = \mathbf{-3.0}$ (much worse than guessing).

The crucial subtlety — R^2 depends on the test set's spread. Because SS_{tot} grows with the test set's variance, the *same model* can score very different R^2 on test sets of different spread. So we **always** report MAE and the test-set standard deviation $SD(y)$ next to R^2 . A very useful identity we use as a self-check (it follows from the definitions when SD uses the population form):

$$R^2 \approx 1 - \left(\frac{RMSE}{SD(y)} \right)^2$$

Worked example (the identity). Our honest split has $RMSE = 95.5$ and $SD(y) = 108.1$, so

$1 - (95.5/108.1)^2 = 1 - 0.780 = \mathbf{0.220}$ — exactly the reported R^2 . Every results row reconciles this way; that is one of our built-in correctness checks (Section 7.3).

r^2 (Pearson) — a looser cousin, reported for honesty. This is the *squared correlation coefficient*: it measures only whether the model's ranking lines up with the truth, ignoring bias and scale, so it is **always** $\geq R^2$. Some papers quietly call this " R^2 ." We print **both** so nobody can quote the looser one against us.

Trap. If a reported " R^2 " looks suspiciously high, check whether it is really squared correlation.

The strict coefficient of determination (ours) penalises systematic bias and wrong scale; squared correlation forgives both.

Spearman (rank correlation). "*Does the model order photos correctly from cleanest to dirtiest?*" 1.0 = perfect ordering. A **high Spearman with a modest R^2** means the model sees pollution (ranks it well) but is off on the exact number — which is exactly our situation (Spearman 0.65, R^2 0.22–0.39).

Coverage & interval width (for C1, Section 6). For the low–high range: **coverage** = how often the truth actually lands inside the range (target 0.90); **width** = the average size of the range. Coverage is meaningless without width (an infinitely wide range always "covers"), so we always report both.

3. The data, and the honesty problem that shapes everything

3.1 The dataset

PM25Vision: street-level photos, each labelled with that location's **daily-average PM2.5 AQI**. After removing 123 duplicate image IDs we use **11,096 photos across 3,259 monitoring stations** (median 2 photos per station, max 91). The imagery comes from Mapillary; the labels from the World Air Quality Index (WAQI) project. The label distribution is **right-skewed** — many moderate-pollution photos, few extreme ones — which matters for training (Section 5.4) and for the accuracy metric.

3.2 Auditing the data before modelling

A modelling result is only as trustworthy as the data under it, so we run three checks first.

- **Exact duplicates.** 123 rows shared an **image_id**; we drop them so the same photo cannot sit on both

sides of a split.

- **Near-duplicates (perceptual hash).** A **perceptual hash** turns each image into a short fingerprint

such that *visually similar images get similar fingerprints* (unlike a cryptographic hash, where one changed pixel changes everything). We group near-identical photos this way. This matters because near-duplicates from the same place are a leakage risk (Section 3.3).

- **The physics falsification test (the clever one).** Before trusting that images and labels are

correctly paired, we compute a simple "how hazy is this photo" number — the mean **transmission** (Section 5.1) — and correlate it with the AQI label across the whole dataset. Physics predicts **hazier photo → more pollution → higher AQI**, i.e. a **negative** correlation between transmission and AQI. We measured **Pearson $r \approx -0.14$** (highly significant, $p \approx 5e-6$). It is a weak-but-real negative correlation in the predicted direction — so the pairing is correct, and the visual signal is real but *subtle*, which foreshadows why the honest R^2 is modest rather than near-zero or near-one.

Key idea. We *falsification-tested* the data: if images and labels were mispaired, haziness would be unrelated to AQI. It correlates negatively exactly as physics predicts, so the data is sound.

3.3 Data leakage — the trap the whole project is built around

Leakage = information from the test set sneaking into training, so the score measures **memorisation** rather than understanding.

Everyday analogy. If a few real exam questions leak into your practice set, your mock score looks great — but it measures memorising those answers, not understanding the subject. On the real exam (new questions), you fall apart.

Why THIS data leaks so easily. Many photos come from the **same monitoring station**, and all photos from one station on one day share a **near-identical daily-average label**. If a random split puts some of a station's photos in training and others in test, the model can score well by **recognising the place** ("I've seen this exact street corner; its label is ~120") instead of **reading the haze**. That is leakage, and our evaluation is specifically designed to forbid it — and then to *measure* how big it is.

3.4 The split family (how we divide photos into train / calibration / test)

We use a **0.65 / 0.15 / 0.20** split and build several **strategies**, changing only *how photos are assigned* so any score difference is purely the split:

- **station-grouped (our primary, honest protocol)**: every station's photos go entirely into one of

train/cal/test. No station straddles the boundary, so the model is tested on ****genuinely unseen places****. This measures real generalisation.

- **random (the leaky control we construct)**: photos assigned independently → same-station photos land

on both sides. This is the shortcut that leaks; we build it deliberately to measure the inflation.

- **geographic**: whole lat/lon regions held out (the hardest test — did it learn haze, or local

scenery?).

- **temporal**: train on earlier dates, test on later ones.

- **shipped**: the dataset's own 80/20 split (which happens to be station-disjoint), used as a

station-disjoint reference.

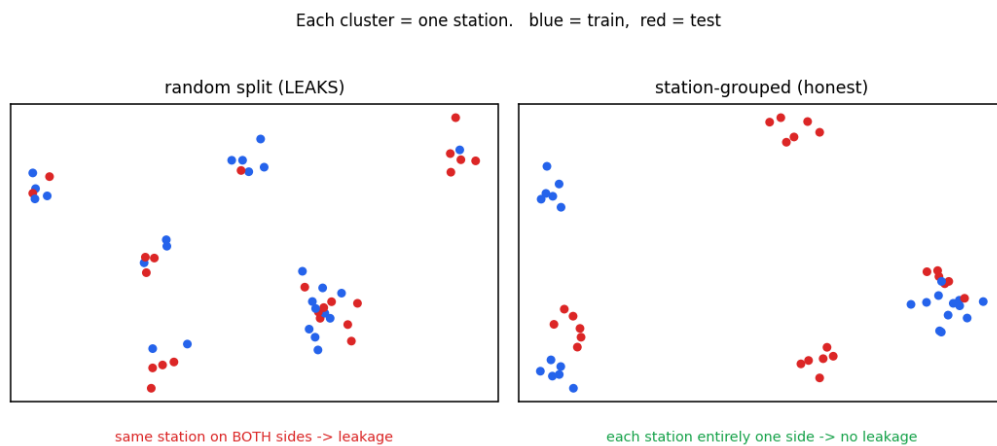


Figure 4. Random vs station-grouped splits (Section 3.4).

The calibration set is held strictly separate from train and test, because the coverage guarantee in Section 6 depends on the calibration data never being seen during training.

4. The decisive experiment — measuring the leakage (the headline result)

Here is the project's central finding. We train the **same model with the same random seed** and change **only the split strategy**, so any difference in score is purely the split — a clean controlled experiment.

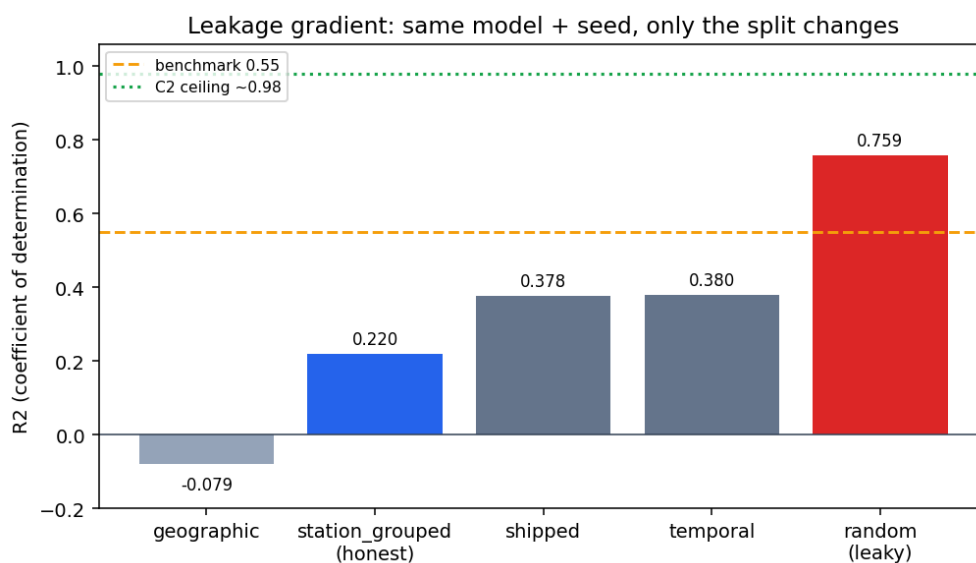


Figure 3. The leakage gradient (Section 4).

4.1 The leakage gradient

split	R ²	MAE	RMSE	SD(y_test)	stations in >1 split
random (leaky control)	0.759	26.4	42.9	87.4	1,221
temporal	0.380	43.5	58.7	74.5	393
shipped	0.378	44.5	64.3	81.5	0
station_grouped (honest)	0.220	54.2	95.5	108.1	0
geographic (hardest)	-0.079	59.9	104.5	100.6	0

Row-by-row reading:

- **random = 0.759.** Deliberately leaky (1,221 stations appear on both sides). It scores ****higher** than

the benchmark's reported 0.550**, and higher than our *entire* honest cross-validated range (Section 7). A leaky split manufactures a big score out of memorised places.

- **temporal = 0.380, shipped = 0.378.** Station-disjoint (or nearly), landing near our honest

cross-validated mean.

- **station_grouped = 0.220.** The honest single number — and, we'll learn, a *hard, high-variance draw*.

- **geographic** = **-0.079**. Holding out whole regions is so hard the model does slightly *worse than

guessing* — an honest measure of how badly single-photo AQI transfers across the globe.

4.2 The leakage gap (Δ_{leak})

- **Lead with the variance-free MAE gap:** $\Delta_{\text{leak}}(\text{MAE}) = 54.2 - 26.4 = \mathbf{27.8 \text{ AQI}}$. The leaky split

roughly **halves the error**. We lead with MAE because, unlike R^2 , it does **not** depend on the test set's spread — so it is the clean, honest measure of the effect.

- $\Delta_{\text{leak}}(R^2)$ is 0.539 against the single hard draw (0.220), or 0.374 against the robust cross-validated

mean (0.385). We report the pair, not just the most dramatic number.

4.3 The causal clincher — contaminated vs clean (no extra training)

A skeptic could object: *"maybe the random test set is just easier."* We rule that out ****within the random model's own test set — same model, same test photos — by splitting them into contaminated**** (their station or a near-duplicate is *also* in that model's training set) vs **clean** (genuinely unseen):

subset	n	R^2	MAE
contaminated (leaked)	1,757	0.762	25.4
clean (no leakage)	463	-0.86	30.4

The high score lives **entirely in the leaked photos**. On genuinely unseen photos the same model is worse than guessing. Because the test set and model are held fixed and only the leakage differs, this is **causal evidence** that the inflation is leakage, not an easier test.

Trap — read the -0.86 correctly (a professor WILL probe this). It is ****variance-amplified**, not a catastrophic failure.** The clean subset is small (463 photos) and dominated by low-variance stations, so its $\text{SD}(y)$ is small; since $R^2 = 1 - (\text{RMSE}/\text{SD})^2$, a *modest* error on a low-spread set drives R^2 sharply negative. Look at MAE: **25.4 vs 30.4** — only a little worse. ****Lead with the MAE gap and the direction (contaminated ■ clean); treat -0.86 as qualitative, not a precise honest estimate.****

Worked example (why small SD makes R^2 explode). Suppose clean truths have $\text{SD} = 20$ and the model's RMSE on them is 30. Then $R^2 \approx 1 - (30/20)^2 = 1 - 2.25 = \mathbf{-1.25}$ — very negative, from an RMSE (30 AQI) that is not even large. Same RMSE on a set with $\text{SD} = 100$ gives $R^2 \approx 1 - 0.09 = \mathbf{+0.91}$. The R^2 number is dominated by the denominator, which is why we lead with MAE.

4.4 What we claim, and what we do NOT claim

We do **not** accuse the benchmark authors of cheating, and we do **not** claim 0.55 is impossible honestly (one honest cross-validation fold reaches 0.574, Section 7). We claim two things, and prove them: (1) the data is **dramatically leakage-prone** (a leaky split inflates past 0.55, to 0.759), and (2) single numbers on this task are **unreliable** (honest R^2 swings 0.15–0.57). The benchmark's *unspecified* split merely **admits** the leakage mechanism.

Key idea (the whole thesis in one breath). Same model, only the split changes: leaky 0.759, honest -0.385; the advantage lives entirely in leaked photos (contaminated 0.762 vs clean -0.86). So the data

leaks dramatically and single numbers are unreliable — that is the finding, honestly framed.

5. Building the honest model (in depth)

Now the machinery that produces the honest predictions. The guiding choice: ****feed the network physics, not just pixels.****

5.1 Physics features — teaching the model what haze looks like

Rather than hope the CNN rediscovers atmospheric optics from raw pixels, we hand it two pre-computed maps that behave the same way in every city. The physical model of a hazy image (Koschmieder / He et al.) is:

$$I(x) = J(x) t(x) + A (1 - t(x))$$

where **I** is the observed pixel, **J** is the true (haze-free) scene, **A** is the atmospheric light (the haze colour), and **t(x) ∈ [0,1]** is the **transmission** (how much light survives to the camera). **t = 1** means no haze; **t → 0** means dense haze washing everything toward **A**.

- **Transmission map (Dark Channel Prior, He et al. 2011).** In a clear outdoor photo, almost every tiny

patch has at least one very dark pixel in some colour channel (a shadow, a dark window, tree bark). Haze adds a pale grey veil that **lifts** those dark pixels. So *how dark the darkest local pixel still is* tells you how much haze sits between camera and scene. We compute the **dark channel** (the per-pixel minimum over colour channels, then a local minimum-filter over a 15×15 patch), estimate the atmospheric light **A** from the haziest/brightest pixels, and then the transmission:

$$t(x) = 1 - \omega \min_{c \in \{R, G, B\}} \left(\min_{y \in \Omega(x)} \frac{I^c(y)}{A^c} \right), \quad \omega = 0.95$$

($\Omega(x)$ is the 15×15 patch around pixel **x**; $\omega < 1$ keeps a little haze so distant scenes still look natural.) Result: **transmission ∈ [0,1]**, **LOW** where haze is dense, **HIGH** in clear air.

- **Inverted-saturation map (Fang et al. 2024).** The dark-channel trick **fails on sky** (sky has nothing

dark in it). But haze also **washes colour out** (mixes in white), lowering **saturation** (colour vividness). Sky has colour to lose, so an *inverted* saturation map (HIGH where saturation is LOW) stays informative exactly where the dark channel breaks down. The two maps cover each other's blind spots.

- **The 5-channel input.** We stack [**R, G, B, transmission, inverted-saturation**] into a 5-channel

image. The RGB channels are normalised with ImageNet statistics (so the pretrained backbone sees what it expects); the two physics channels are standardised to a similar scale (otherwise their small [0,1] variance would make the pretrained stem down-weight them). These maps are **cached to disk once** — recomputing the Dark Channel Prior every training epoch would be far too slow (~1.1 GB cache for the full dataset).

Key idea. Low transmission = dense haze; the inverted-saturation map catches haze in the sky where the dark channel can't. We give the network these two maps so it reads haze directly.

5.2 The backbone and the 3→5 channel stem

We use **EfficientNet-B0**, pretrained on ImageNet, but our input has **5** channels, not the usual 3. We **widen the first convolutional layer** from 3 to 5 input channels; the **timm** library initialises the two new channels from the pretrained RGB weights (a routine engineering step, not a research risk). We ask the backbone for its **pooled feature vector** (by setting **num_classes=0**), not a classification — we attach our own prediction heads to that vector. (Those same features later power the abstention OOD gate, Section 8.2.)

5.3 The interval heads — three quantiles that can never cross

A **quantile** answers "what value is this fraction of outcomes below?" The 95th percentile is a value the truth is below 95% of the time; the 5th percentile is a value it is above 95% of the time; the 50th is the median. We predict the **5th, 50th, and 95th percentiles** of the AQI so the 5th–95th pair forms a low–high interval.

Worked example (quantiles). For the numbers [10, 20, 30, 40, 100], the 50th percentile (median) is 30, the "low" (5th-ish) is near 10, and the "high" (95th-ish) is near 100. Notice the median (30) sits *below* the mean (40) because the 100 pulls the mean up — this skew fact returns in Section 5.4.

If we predicted the three percentiles freely, the model could output a 95th **below** its 50th — nonsense. We prevent that **by construction**: predict the lowest quantile freely, then add strictly-positive **softplus** increments and cumulatively sum them:

$$q_1 = z_1, \quad q_k = q_{k-1} + \text{softplus}(z_k) \quad (k > 1), \quad \text{softplus}(z) = \log(1 + e^z) > 0$$

Because every increment is strictly positive, the outputs can only increase $\rightarrow \mathbf{q05 \leq q50 \leq q95 \text{ always}}$, with no penalty term and no post-hoc sorting. The heads work in **log-AQI** space (stable on a skewed target); exponentiating back to AQI preserves the ordering (exp is monotone).

5.4 The mean-vs-median problem, and the dedicated point head

This is a subtle but important design decision, and a favourite professor question.

- On a **right-skewed** target (few very-high-AQI photos), the **median sits below the mean** — the long

high tail pulls the mean up but not the median.

- The **median minimises absolute error**, so the 50th-percentile head naturally predicts the median.
- But **R² rewards matching the conditional mean**, not the median (R² is built on squared error, whose

minimiser is the mean). So reporting the median would **structurally depress R²** even if the model is good.

Right-skewed labels: median sits BELOW the mean (why we use an MSE point head)

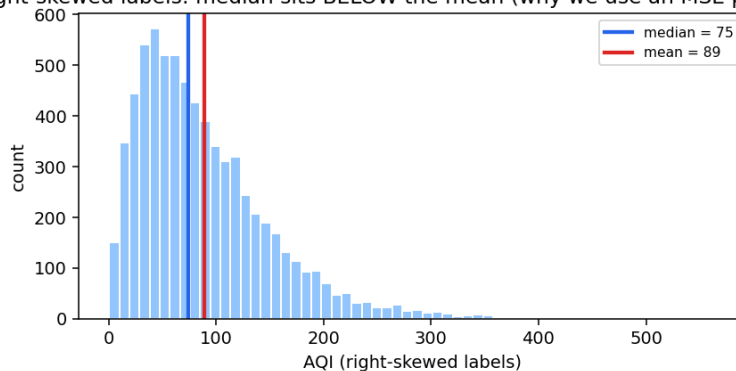


Figure 5. Why the median is below the mean on skewed labels (Section 5.4).

Our fix: a **separate point head** trained with **MSE (squared error) on a standardised target**. MSE's minimiser is the mean — exactly what R² rewards. We standardise the target so a fresh head starts by predicting the dataset mean (a good starting point) and there is no unstable retransformation:

$$\tilde{y} = \frac{y - \mu}{\sigma} \longrightarrow \hat{y}_{\text{AQI}} = \sigma \hat{\tilde{y}} + \mu$$

where μ, σ are the **train-set** AQI mean and standard deviation (stored inside the model so they travel with the checkpoint). At inference we de-standardise linearly back to AQI. **The point head is our accuracy output (MAE/RMSE/R²); the quantile heads still give the interval.**

Trap. An earlier version mistakenly trained this head with a robust (Huber) loss whose small delta made it learn the **median** — which crushed R². Switching to **MSE on the standardised target** was the fix. Say: "point head = MSE + standardise + isotonic — mean-seeking; not the median, not Huber."

5.5 The losses

Pinball (quantile) loss teaches each head its percentile with an **asymmetric** penalty:

$$L_{\tau}(y, \hat{y}) = \max(\tau(y - \hat{y}), (\tau - 1)(y - \hat{y}))$$

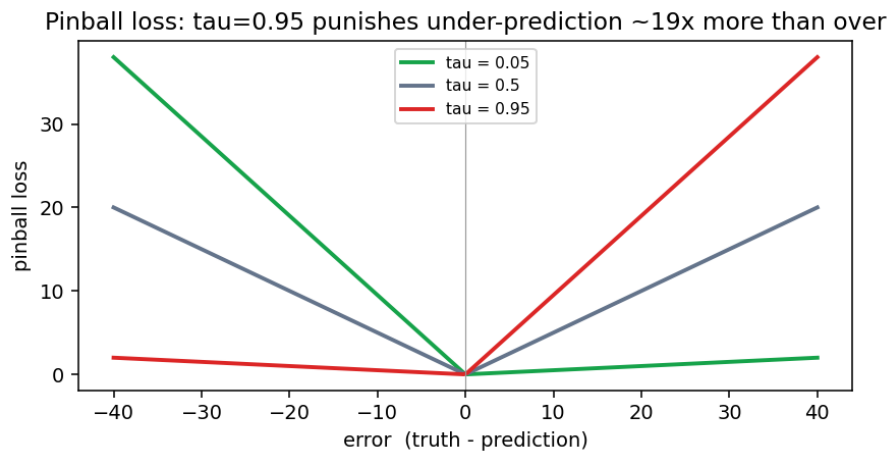


Figure 6. The asymmetric pinball loss (Section 5.5).

- For $\tau = 0.95$, under-prediction (guessing too low, so $y - \hat{y} > 0$) is penalised with weight 0.95,

while over-prediction is penalised with weight $|0.95 - 1| = 0.05$. The ratio is $0.95/0.05 = 19\times$ — under-shooting the 95th percentile hurts 19 times more than over-shooting, so that head learns to sit near the *top* of the plausible range.

- For $\tau = 0.05$ it is the mirror (over-prediction hurts 19x), so that head sits near the bottom.
- For $\tau = 0.50$ the penalty is symmetric and this reduces to ordinary absolute error (the median).

Worked example (pinball asymmetry). Truth $y = 100$, $\tau = 0.95$. If the head predicts 80 (under by 20): loss = $\max(0.95 \cdot 20, -0.05 \cdot 20) = 19$. If it predicts 120 (over by 20): loss = $\max(0.95 \cdot (-20), -0.05 \cdot (-20)) = \max(-19, 1) = 1$. Same 20-point error, but under-shooting costs 19x more — that is how the loss pushes the 95th-percentile head upward.

Point loss (MSE) on the standardised target, as in 5.4. The total training loss is **pinball(quantile heads) + weight · MSE(point head)**.

5.6 Training

AdamW optimiser (learning rate $3e-4$, weight decay $1e-4$), a **cosine-annealed** learning rate with a 2-epoch warm-up (start slow, ramp up, then smoothly decay — helps stability and final accuracy), batch size 32, up to 60 epochs with **early stopping on the calibration point-MAE** (stop when the metric we care about stops improving, and keep the *best* checkpoint, not the last), mixed precision (faster on GPU), fixed seeds (reproducibility). Two small-data helpers:

- **Class-balanced sampling** (train loader only): a weighted sampler oversamples the rare high-AQI photos

so the model actually sees the extremes; calibration and test keep the **natural** distribution so the guarantee and the reported numbers stay honest.

- **Stochastic depth / drop-path**: randomly skips residual branches during training — a regulariser that

helps on our modest dataset.

We also implemented three optional **accuracy levers**, all selected on calibration and all honest: **EMA** (an exponential moving average of the weights, usually a bit more accurate and stable), **test-time augmentation** (average the prediction over the photo and its mirror image), and a ****seed ensemble**** (train several models with different seeds and average their outputs).

6. Honest uncertainty — turning predictions into a *guaranteed* range (C1)

A raw model can be over-confident (interval too narrow) or under-confident (too wide). We fix that with two post-processing steps that need no retraining.

6.1 Conformalized Quantile Regression (CQR) — the coverage guarantee

CQR gives a **distribution-free, finite-sample guarantee**: the true value lands inside the interval $\geq 90\%$ of the time, for **any** model (a weak model just gets honestly-wide intervals), as long as calibration and test data are **exchangeable** (drawn from the same distribution). The recipe (Romano, Patterson & Candès, 2019):

Step 1. On the **calibration** set, record how far outside its predicted range each truth fell:

$$E_i = \max(\hat{q}_{05}(x_i) - y_i, y_i - \hat{q}_{95}(x_i))$$

E_i is negative when the truth was comfortably inside the interval, and positive by exactly the size of the miss when it fell outside.

Step 2. Take Q = the finite-sample $(1-\alpha)$ quantile of those scores ($\alpha = 0.10$ for 90% coverage):

$$Q = \text{quantile}\left(\{E_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$$

(The **(n+1)** correction is what makes the guarantee exact for a finite calibration set rather than only asymptotically.)

Step 3. Widen **every** future interval by Q at both ends, and clip the lower end at 0 (AQI can't be negative): $[\text{q05} - Q, \text{q95} + Q]$.

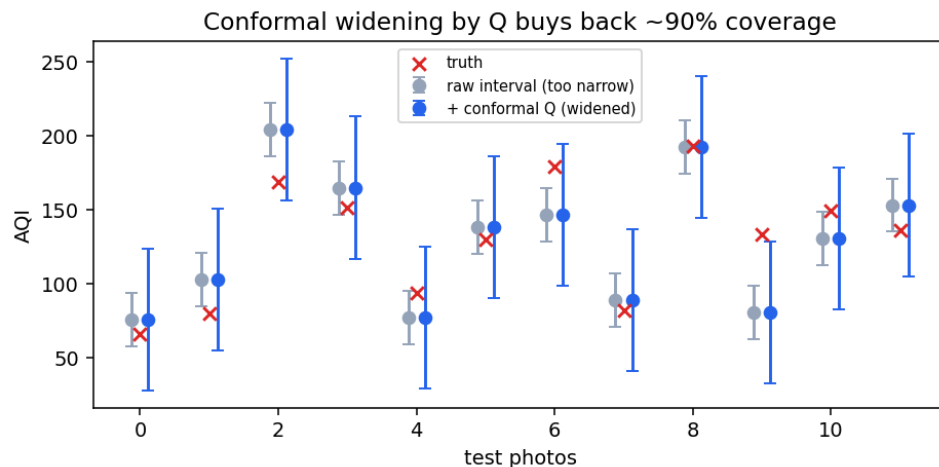


Figure 7. Conformal interval widening (Section 6.1).

Why it works, intuitively: Q is your model's **own typical "how much I miss by"**, measured on held-out data. Padding every interval by that amount buys back exactly the coverage you were missing. It is one number and it is model-agnostic.

Worked example (conformal Q). Say the calibration miss-scores E_i , sorted, are $[-40, -22, -10, -3, 5, 18]$ ($n = 6$). For 90% coverage the level is $\lceil 7 \cdot 0.9 \rceil / 6 = \lceil 6.3 \rceil / 6 = 7/6 \rightarrow$ clipped to the top value $\rightarrow Q = 18$. Every future interval is widened by 18 AQI at each end. Empirically we then verify coverage ≈ 0.90 .

What is exchangeability, and can it break? Exchangeability means calibration and test points are "interchangeable" draws from one distribution. Testing on **unseen stations** mildly breaks this (new places differ a little from calibration places), so our single-split coverage drifts to 0.87 — slightly below 0.90. This is honest, expected drift, and cross-validation coverage is 0.895 (Section 7.2).

6.2 Isotonic recalibration — removing systematic bias

The point estimate tends to **under-predict** the rare high-AQI photos, and that systematic bias hurts R^2 . **Isotonic regression** learns a **monotonic** (non-decreasing) map from prediction $\rightarrow$ truth, fit on the **calibration** set only, and applied to test (never fit on test). It says e.g. "when the model outputs ~ 260 , the truth averages ~ 340 , so map $260 \rightarrow 340$ ", removing the bias while preserving the ordering (so it cannot scramble the ranking). It is a standard, low-risk post-hoc step and needs no retraining.

Worked example (isotonic). Suppose on the calibration set, whenever the raw head outputs ~ 200 the truth averages 210, and whenever it outputs ~ 300 the truth averages 360 — the head under-predicts, and more so at the top. Isotonic learns the non-decreasing map $200 \rightarrow 210$, $300 \rightarrow 360$ (interpolating between). At test, a raw 260 is mapped up to roughly 285. Because the map only ever *increases* with the input, two photos the model ranked in a given order stay in that order — so it fixes the *level* without touching the *ranking*. That is why it can lift R^2 and MAE without risk of scrambling Spearman.

6.3 Coverage vs width

We always report the two together. Our honest single split gives coverage ≈ 0.87 (the exchangeability drift above); cross-validation gives mean coverage **0.895 $\approx$ 0.90**, which **validates** the whole calibration machinery across independent folds.

7. How good is good? — evaluation and rigor

7.1 The honest single-split result

On the station-grouped (leakage-safe) split: $R^2 = 0.220$, MAE 54.2, RMSE 95.5, Spearman 0.649, coverage 0.872, SD(y) 108.1 (**n_train 7,217, n_test 2,216**). **Reading: the model** ranks** pollution well (Spearman 0.65) and its intervals are near-calibrated, but its point- R^2 is modest — and this particular split, as cross-validation reveals, was a **hard, high-variance draw**.

7.2 Cross-validation — is that 0.22 trustworthy?

A single split could be lucky or unlucky. **Cross-validation** re-runs the whole evaluation on several different splits and reports the spread. But **ordinary k-fold would leak here** (it shuffles all photos, re-mixing same-station photos across train/test — exactly the flaw we expose). So we use **GroupKFold** on **station_id**: the stations are divided into 5 groups; each fold tests one group and trains on the other four, so **every station is tested exactly once and never appears in its own fold's training data** (0 straddling). Within each fold's training stations we carve a station-grouped calibration set, keeping the 0.65/0.15/0.20 shape.

Result (5 folds):

fold	0	1	2	3	4
R^2	0.405	0.574	0.365	0.429	0.152

Honest $R^2 = 0.385$ (SD across folds 0.15; 95% CI [0.196, 0.574]) · MAE 46.3 · Spearman 0.679 · **coverage 0.895**.

Three things this tells us:

1. **The robust honest number is 0.385 — higher than the single split's 0.220**, which turns out to be one

hard, high-variance draw ($\approx$ fold 4's 0.152). Cross-validation *raised* our headline.

1. **The conformal intervals are validated** — mean coverage 0.895 $\approx$ 0.90 across five independent folds.
2. **Honest R^2 is highly split-dependent (0.15–0.57)**. A *single* R^2 on this task is unreliable — which

is exactly why single-number benchmarks (including 0.55) must be distrusted.

The honest correction this forces: one fold reached **0.574 > 0.55**, so we retract any claim that "0.55 is unreachable honestly." 0.55 is *within* the honest range; the leakage proof is the causal test (§4.3) and the leaky 0.759 exceeding the whole CI, **not** the impossibility of 0.55.

On the $\pm$ and the CI (a professor may probe this). "95% CI [0.196, 0.574]" is a Student-t interval with half-width $\approx$ 0.19; the plain standard deviation across folds is 0.15 (a *different* quantity — do not confuse them). And these 5 folds come from **one** deterministic GroupKFold partition, so they are **correlated** (they share training data), which makes the plain t-interval mildly **optimistic**. A fully rigorous version repeats grouped CV over several seeds with the Nadeau–Bengio correction. The qualitative story (0.385, split-dependent, validated coverage) is unchanged.

7.3 How we know none of this is a bug — two correctness audits

Twice (2026-09-25 and 2026-09-28) we ran **three independent read-only audit agents** over every module. Combined verdict: **the accuracy numbers, the leakage gradient, and the causal result are computed correctly and honestly** — no

result-affecting bug.** They confirmed: prediction $\leftrightarrow$ label $\leftrightarrow$ row alignment (so the contaminated/clean split is meaningful); station-disjointness holds *by construction*; the conformal math is correct (a simulation hit coverage 0.9006); the point head is genuinely mean-seeking; and **every results row reconciles** with the identity $R^2 \approx 1 - (\text{RMSE}/\text{SD})^2$ (e.g. $1 - (95.5/108.1)^2 = 0.220$). The one over-statement they caught was C2's first run (Section 8.1), now fixed.

8. The two extra contributions

8.1 C2 — the unavoidable-error ceiling

The idea in one line: you can't score full marks on an exam graded with a smudged answer key — and our answer key (daily-average AQI) is partly smudged for an instantaneous photo. So there is a hard ****upper bound**** on achievable R^2 .

The formula. Write $\mathbf{y}$ for the daily-average label and $\mathbf{y}^*$ for the instantaneous truth a perfect photo-reader would output. The residual it still suffers is the within-day deviation $\epsilon = \mathbf{y}^* - \mathbf{y}$, with variance $\text{Var}(\epsilon)$. The best possible R^2 on predicting the labels is then:

$$R_{\max}^2 = 1 - \frac{\text{Var}(\epsilon)}{\text{Var}(y)}$$

where $\text{Var}(y)$ is the spread of the labels **on the evaluation split** and $\text{Var}(\epsilon)$ is the typical **within-day variance** of AQI.

How we estimate $\text{Var}(\epsilon)$ — three careful refinements:

1. **Per-hour InstantCast conversion.** We pull **hourly** PM2.5 from **OpenAQ** (a free global

air-quality API), convert *each hourly reading* to AQI, and take the variance of those hourly AQIs about the day's mean — matching how the labels are built (not NowCast, not variance in $\mu\text{g}/\text{m}^3$ space).

1. **Level-reweighting (the load-bearing fix).** Within-day swing depends on the pollution level, so a

single average is misleading if the reference stations have a different pollution mix than our photos. We estimate the **noise curve** $v(m) = E[\text{Var}(\epsilon) \mid \text{day-mean AQI band}]$ and reweight it to PM25Vision's *own* label histogram $p(m)$:

$$\text{Var}(\epsilon) = \sum_m p(m) v(m)$$

1. **Split-specific $\text{Var}(y)$.** The ceiling scales with the test set's spread, so we use the station-grouped

test split's label variance (SD ≈ 108 , so $\text{Var}(y) \approx 11,708$).

The measured result (32 OpenAQ stations, 6 countries, 1,241 station-days): $\text{Var}(\epsilon) \approx 208.6$ (optimistic) to 247.9 (conservative), $\text{Var}(y) \approx 11,708$, giving **$R^2_{\max} \approx 0.98$** — specifically 0.982 (optimistic) to **0.979** (conservative, imputing the uncovered high bands), and 0.980 under the 2024 EPA table (**breakpoint-insensitive**). The **interval-width floor** is ≈ 35 AQI (no honest 90% interval should be narrower than the pollution's own within-day spread).

Worked arithmetic (the ceiling). $1 - 208.6/11,708 = 1 - 0.0178 = \mathbf{0.982}$ (optimistic).

$1 - 247.9/11,708 = 1 - 0.0212 = \mathbf{0.979}$ (conservative). Both round to ~ 0.98 .

Worked example (level-reweighting). Using the $v(m)$ curve and dataset weights below:

$\text{Var}(\epsilon) = 0.245 \cdot 129.4 + 0.255 \cdot 104.7 + 0.246 \cdot 501.8 + 0.254 \cdot 104.9 \approx 31.7 + 26.7 + 123.4 + 26.6 =$

$\approx \mathbf{208}$ — the optimistic $\text{Var}(\epsilon)$. Notice the 100–150 band contributes the most (its within-day swing,

$v = 501.8$, is by far the largest), which is exactly why *missing* the even-higher bands would bias the

ceiling; the conservative estimate fills them with a high v and still only reaches $\text{Var}(\epsilon) \approx 248$.

The $v(m)$ curve (within-day AQI variance by pollution band):

band (AQI)	station-days	v (variance)	within-day SD	dataset weight
0–50	715	129.4	11.4	0.245
50–100	430	104.7	10.2	0.255
100–150	86	501.8	22.4	0.246
150–200	10	104.9	10.2	0.254

What it means (the payoff): label noise is only ~2% of the label variance ($\text{Var}(\epsilon) \approx 208\text{--}248$ vs $\text{Var}(y) \approx 11,708$). So daily-average labeling costs at most ~2 points of R^2 , and our honest $0.22 / 0.385$ sits far below the ~0.98 ceiling. **Therefore the gap is the** difficulty of reading pollution from one photo (the visual task), NOT noisy labels.** That is exactly what C2 exists to establish: the ceiling's job was to answer "*is the remaining gap difficulty or bad labels?*" — and the answer is emphatically **difficulty**.

Trap — own these caveats before the professor states them. (i) The first pass used only ~13 stations and gave a *looser* 0.977; we feared the true ceiling might drop to ~0.90–0.95 once high-AQI bands were covered. The proper 32-station re-run shows it does **not** (holds at ~0.98). (ii) The reference still under-covers the very-high-AQI bands (200+, ~13% of label mass), which are **imputed conservatively**; even so the ceiling only drops to 0.979. (iii) Report it as **~0.98, an upper bound** — not a single decimal, and not with a tight CI (the bootstrap band comes from few stations). (iv) The ceiling is **split-specific**, so compare a split's R^2 only to *its own* ceiling — we prove leakage with the causal test, not a " $R^2 > \text{ceiling}$ " overlay.

8.2 C3 — inference-time abstention (knowing when to refuse)

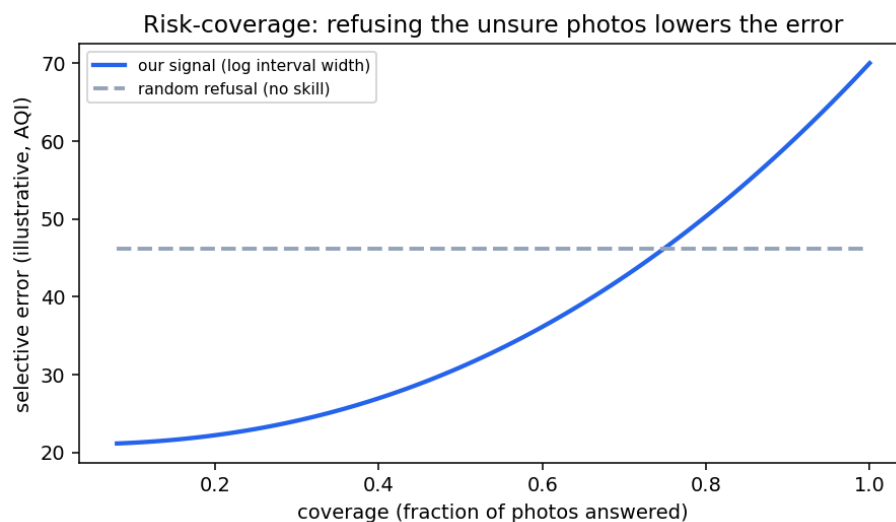


Figure 8. Risk-coverage schematic for abstention (Section 8.2).

An honest system must sometimes say "**I can't answer this.**" Two **independent** gates, checked in a deliberate order:

Gate 1 — OOD / unusable (checked FIRST): "is this even a valid daytime street photo?" Night, indoor, and blank-texture photos are **out of distribution (OOD)** — the model was never trained on anything like them, so any number it produces is meaningless. Our gate uses **interpretable, station-invariant checks**: a photo is flagged if it is **too dark** (night) or **too flat / featureless** (near-zero detail). These are the right signal because a valid daytime street photo from *any* station is bright and detailed, so the gate does not punish unseen locations. **One honest caveat:** a well-lit **indoor** scene is bright *and*

detailed, so these checks do **not** catch it — indoor is a documented near-OOD limitation of the deployed gate (the Mahalanobis detector below would catch indoor, but at the cost of over-refusing valid unseen-station outdoor photos — so neither is free, and we choose the station-robust one).

We also *tried* the textbook approach — a shape-aware **Mahalanobis distance** in the backbone feature space —

$$D_M(f) = \sqrt{(f - \mu)^\top \Sigma^{-1} (f - \mu)}$$

where $\mathbf{f}$ is the photo's feature vector and μ, Σ are the mean and (Ledoit–Wolf-shrunk) covariance of the good-image features; it measures *how unusual* a photo is relative to real street photos. ****But we found it over-refuses valid photos from unseen stations**** (it refused ~99% of good test photos in one run): in ~1280-dim feature space, a new station's photos are genuinely far from the calibration photos, so the detector confuses "new place" (benign) with "unusable photo". That is an honest, instructive finding — a raw feature-distance detector is the wrong tool for a *station-generalisation* task — so we report its AUROC for comparison but use the robust handcrafted gate. (We deliberately do **NOT** use interval width for OOD either — the conformal guarantee is void off-distribution, so a width-based test would be meaningless.)

Gate 2 — uncertainty (checked second): "the photo is valid, but is the model too unsure?" We refuse the valid photos with the widest **log-space** interval width, **q95_log – q05_log**. Log space matters: because we train on $\log(\text{AQI})$, the raw-AQI width grows with pollution level, so a raw-width rule would just refuse every high-AQI photo — the log width is the level-fair uncertainty signal.

Thresholds are set on good images only (never on labels or the OOD sets). For τ_{ood} we split the held-out unseen-station good photos into **two disjoint slices**: we *fit* the 5% false-refusal budget on one (**good_fit**) and *report* the achieved refuse-rate on the other (**good_eval**) — otherwise "≈5% refused" would be true by construction rather than a real check. The threshold uses an exact **top-k rule** (refuse the $\lceil 5\% \cdot n \rceil$ highest-scoring good photos) instead of a raw percentile, because the handcrafted score is spiked at exactly 0 for most good photos and a plain percentile could collapse to 0 and refuse arbitrary mildly-dim photos. τ_{width} = the calibration log-width at the target coverage. The gate logic: if **ood_score** > τ_{ood} → abstain (OOD); else if **log_width** > τ_{width} → abstain (uncertain); else answer.

What we measure. Selective prediction via **risk–coverage** curves (accept the most-confident fraction, watch the error), the **AURC** (area under the risk–coverage curve — lower is better), and selective MAE/RMSE/category/coverage at 90/80/70% coverage; plus baselines (random, oracle). We report these for the **deployed log-width signal** as the headline (report == deploy), with the raw-AQI width shown only as a labelled diagnostic — and we state the honest finding that on this hard task log-width is a *weak* error predictor, so the OOD gate, not the uncertainty gate, is C3's main practical value. For the OOD gate, **OOD-detection** metrics against three external "should-refuse" sets pushed through the *identical* physics pipeline — **ExDark** (night, far-OOD), **DTD** textures (featureless, far-OOD), and **MIT-Indoor** (indoor, near-OOD): **AUROC** (ranking quality, 0.5 = chance, 1.0 = perfect), **AUPR**, and **FPR@95TPR** (the false-positive rate when we catch 95% of the OOD images — lower is better), plus a **refuse-rate table** (high on the *far*-OOD sets, ≈5% on held-out **good_eval**, and — honestly — near chance on MIT-Indoor for the handcrafted gate, the indoor limitation made measurable). To show the Mahalanobis over-refusal is genuine *station shift* and not just in-sample optimism, we set its threshold on a **held-out same-station** slice and report the excess refusal on unseen-station test.

Worked example (AUROC). Give every photo its OOD score (here the handcrafted too-dark/too-flat score). AUROC is the probability that a randomly chosen *unusable* photo scores higher than a randomly chosen *good* photo. If good photos score, say, [1, 2, 3] and night photos score [4, 5, 6], every night photo out-scores every good photo → AUROC = 1.0 (perfect separation). If the two sets overlap completely → AUROC = 0.5 (chance). **FPR@95TPR** asks: to catch 95% of the unusable photos, how many *good* photos do we wrongly refuse? Lower is better. These let us report OOD-detection quality as single, comparable numbers.

(C3 is built and locally tested; the exact metric values land after the **08_abstention.ipynb** run.)

9. Delivering it — the live demo and the full picture

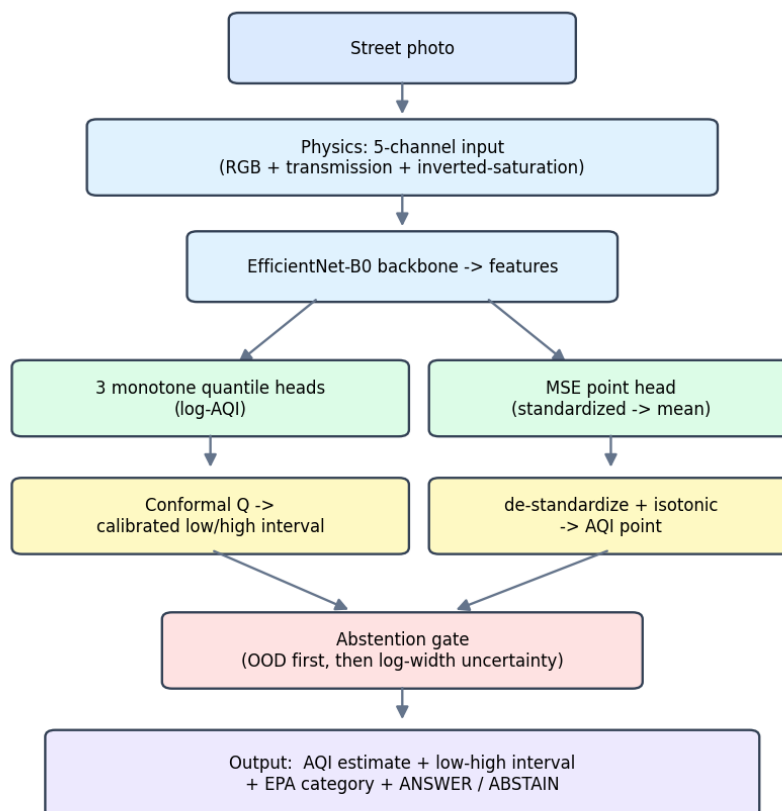
9.1 The one-photo pipeline (the demo)

predict(photo) runs the whole trained pipeline on a single image: build the 5-channel physics input → EfficientNet-B0 → invert the log quantiles → apply the conformal widening Q → isotonic-recalibrate the point estimate → run the abstention gate. It returns the **AQI estimate**, the **low–high interval**, the **EPA category + colour**, an **answer / abstain** decision with a reason, and the two physics maps. A **Gradio** app wraps this, deployed as a permanent **Hugging Face Space** (a free, always-on public web app) — so the honesty is *visible*: upload a night photo and the system tells you it won't guess.

Worked example (one photo, end to end). A daytime street photo comes in. (1) We compute its transmission and inverted-saturation maps and stack them with RGB → a (5, 224, 224) tensor. (2) EfficientNet-B0 turns it into a ~1280-number feature vector. (3) The quantile heads output, in log space, [q05, q50, q95]; exponentiating gives, say, [95, 130, 190] AQI. (4) Conformal widening adds $Q \approx 30$: the interval becomes [65, 220]. (5) The point head outputs a standardized value; de-standardise and isotonic-recalibrate → AQI ≈ 148 . (6) The abstention gate computes the handcrafted OOD score (too-dark/too-flat — here below τ_{ood} → a valid daytime photo) and the log width (below τ_{width} → confident) → **answer**. Output: "AQI ≈ 148 , range 65–220, category *Unhealthy for Sensitive Groups*, ■ answer." A night photo would instead fail the OOD gate at step 6 (too dark) and return "■ abstain — too dark (night)."

9.2 How everything connects (the end-to-end picture)

End-to-end pipeline: every step exists to make the number trustworthy



Upstream (offline): audit data -> leakage-safe split -> train -> cross-validate -> C2 ceiling

Figure 9. The full pipeline, end to end.

Read the pipeline as one honest chain: **audit the data → split it so it can't cheat → turn each photo into physics-aware features → predict a calibrated range and a mean-seeking point → guarantee the range's coverage and de-bias the point → evaluate leakage-safely and cross-validate → bound the achievable accuracy (C2) → refuse when unsure (C3) → serve it live.** Every box exists to make the final number **trustworthy**, not merely high. That is the project's identity.

10. Presenting and defending

10.1 The five-minute story (rehearse out loud)

1. **Goal & honesty (30s).** Estimate AQI from one street photo, honestly — a calibrated range, an error

ceiling, refuse-to-answer, and leakage-free evaluation.

1. **The data & the trap (45s).** ~11k photos, ~3,260 stations, many photos per station sharing one

daily-average label — which makes the data dangerously easy to leak.

1. **The ruler (45s).** Split into train/cal/test; grade with R^2 (how much better than guessing the

average; can be negative), MAE, and Spearman.

1. **The finding (90s).** Same model, only the split changes: honest (cross-validated) **0.385 ± 0.19**

(folds 0.15–0.57), leaky **0.759** — above the whole honest range. Inside the leaky split the advantage is entirely in leaked photos (contaminated 0.762 vs clean –0.86) — proof it's leakage, not an easier test. We don't claim 0.55 is impossible (a fold hit 0.574); the point is single numbers are unreliable and the data leaks dramatically.

1. **The honest machine (60s).** Physics 5-channel input, EfficientNet-B0, a mean-seeking point head for

accuracy, conformal-calibrated intervals for honesty (~90% coverage), isotonic de-biasing.

1. **Bulletproofing (30s).** An error ceiling (~0.98, an upper bound) shows the score is task-limited not

label-limited; leakage-safe cross-validation gives $0.385 \pm \text{CI}$ with validated coverage; two 3-agent audits found no bug.

1. **Contribution (20s).** An honest number **plus** a measured, causally-proven flaw in how these

benchmarks are usually scored — plus abstention and a live demo.

10.2 The professor Q&A bank

Bold = the answer to say; the rest is backup if pushed.

Framing & motivation

- *In one sentence, what did you do?* → ****Estimate AQI from a street photo honestly, and prove a**

benchmark's high score is inflated by leakage.**

- *Why AQI and not $\mu\text{g}/\text{m}^3$?* → **It's the label the dataset ships and what apps show users;** training in AQI

avoids a lossy conversion and stays comparable to the benchmark. $\mu\text{g}/\text{m}^3$ appears only in the C2 ceiling.

- *What's the real contribution — image→AQI already exists?* → **Honesty and measurement:** we expose and

measure leakage inflation (Δ_{leak}), prove it causally, and add calibrated intervals + an error ceiling + cross-validation + abstention.

- *Why should anyone trust a lower number than the benchmark?* → ****Because the higher number is inflated**

by leakage, which we prove; the honest, robust number is 0.385 with validated intervals.**

Metrics

- *What is R^2 ?* → **The fraction of the truth's variation the model explains vs always guessing the mean;**

1 perfect, 0 = mean, negative = worse than the mean.

- *R^2 can be negative?* → **Yes** — geographic split (−0.079) and the clean subset (−0.86) are real

negatives; the 0-to-1 intuition is for *squared correlation*, a different quantity.

- *Why report both R^2 and r^2 (Pearson)?* → **Because papers use " R^2 " for two different things;** the strict

R^2 penalises bias/scale, squared correlation doesn't and is always $\geq$ it. We print both so nobody quotes the looser one against us.

- *Why is 0.385 good when the benchmark got 0.55?* → **It's field-normal and honestly measured;** the

closest honest analogue (Mondal 2024) is $\approx$ 0.39, and 0.55 sits at the optimistic edge, *within* our fold range.

- *Isn't 0.22 just a bad model?* → **No:** Spearman 0.65 (it ranks pollution), near-calibrated intervals,

and it sits far below the $\sim$ 0.98 ceiling — the low point- R^2 is the task's difficulty, not a failure to learn.

- *Why is MAE 46–54 acceptable on a 1–530 scale?* → **The task is genuinely hard (single instant vs

daily-average label) and MAE is variance-free;** it's in the field-normal range and we report it beside R^2 precisely so the picture is honest.

Leakage

- *What is leakage here?* → **Same-station photos share a near-identical label;** a random split lets the

model win by recognising the place, not reading the haze.

- *How did you PROVE it, not just claim it?* → **Two experiments:** the gradient (same model, only the

split changes → 0.759 down to −0.079) and the causal test (contaminated 0.762 vs clean −0.86 — the advantage lives entirely in leaked photos).

- *Couldn't the random test set just be easier?* → **No — contaminated-vs-clean uses the same test set and

model;** only leakage differs and it explains the whole gap.

- *Do you claim the authors used a random split?* → **No; their split is unspecified.** We claim it

admits the mechanism; a leaky split reproduces and exceeds 0.55, and the causal test proves the inflation is leaked photos.

- *Why is clean- R^2 only "qualitative"?* → **It's from the random-trained model on a small, low-variance

subset, so −0.86 is variance-amplified;** the load-bearing claim is the direction and the MAE gap.

Splits & cross-validation

- *Why station-grouped splitting?* → **Because the unit of leakage is the station;** grouping tests the

model on genuinely unseen places.

- *Why GroupKFold, not plain k-fold?* → **Plain k-fold re-leaks same-station photos**; grouping by station

keeps every fold honest. It turns one number (0.22) into $0.385 \pm \text{CI}$.

- *Is your CI rigorous?* → **It's a Student-t CI across 5 folds**, which is mildly optimistic because the

folds are correlated;**the rigorous version is repeated CV with the Nadeau–Bengio correction** — we state this openly.

Model & physics

- *Why physics channels instead of raw pixels?* → **They encode the haze signal explicitly and generalise**

across cities,**so the model doesn't have to rediscover atmospheric optics from pixels.**

- *How does the Dark Channel Prior measure haze?* → **Clear patches have a very dark pixel; haze lifts it;**

so 1 – darkchannel estimates transmission (low = dense haze). The inverted-saturation map catches haze in the sky where the dark channel fails.

- *Why a separate point head, not the median?* → **The median under-shoots on skewed data and depresses**

R^2 ;**the MSE point head estimates the mean, which R^2 rewards.**

- *Why are the quantiles guaranteed not to cross?* → **Cumulative softplus**: the lowest is free, each step

up adds a strictly-positive increment, so outputs can only increase.

- *Why train on $\log(\text{AQI})$?* → **The label is right-skewed**; log stabilises the loss so a few extreme photos

don't dominate, and exponentiating back preserves ordering.

Uncertainty & calibration

- *Why a range instead of a number?* → **Honesty about uncertainty** — and conformal calibration

guarantees ~90% coverage, which a lone number can't.

- *Why does the conformal guarantee hold?* → **Measure your typical miss on held-out calibration data,**

widen every interval by it; under exchangeability this provably gives $\geq 90\%$ coverage for any model.

- *Coverage is 0.87 not 0.90 — is it broken?* → **No** — testing on unseen stations slightly breaks

exchangeability,**so coverage drifts a little; CV coverage is 0.895. Honest, not a bug.**

- *What does isotonic recalibration do?* → **Removes the systematic tail bias** with a monotone

prediction→truth map fit on calibration; it can't reorder predictions, so it's low-risk.

Ceiling & abstention

- *What is the error ceiling and why an upper bound?* → **The best R^2 possible given daily-average labels**

vs instant photos; an upper bound because it counts label noise only (real models also lose to weak signal).

- *Is $R^2_{max} \approx 0.98$ trustworthy?* → **As an upper bound, yes** — 0.979–0.982, breakpoint-insensitive, 32

stations / 6 countries; report ~ 0.98 (not a tight CI), with the top $\sim 13\%$ extrapolated conservatively.

- *What does the ceiling actually prove?* → ****That label noise is tiny ($\sim 2\%$), so our gap to it is the**

visual task, not the labels** — it reframes "our R^2 is low" as "the task is hard, and we're honest about it."

- *How does the model decide to refuse?* → **Two gates:** an OOD gate (interpretable, station-invariant

"too dark / too flat" checks) first, then an uncertainty gate (log interval width); a 5% false-refusal budget on good photos. *(We also tried a feature-space Mahalanobis detector but it over-refuses unseen stations — an honest finding, so we use the robust semantic checks instead.)*

- *Why not use interval width for OOD?* → **Conformal validity is void off-distribution**, so width is

meaningless there; we use feature distance instead.

- *What are AUROC and FPR@95?* → **AUROC = how well the OOD score ranks unusable above usable photos**

(0.5 chance, 1 perfect); **FPR@95 = the false-refusal rate when we catch 95% of the unusable ones** (lower is better).

Delivery & rigor

- *How do you know there are no bugs?* → **Two independent 3-agent audits verified every module**, plus the

$R^2 \approx 1 - (\text{RMSE}/\text{SD})^2$ self-check reconciles every row.

- *What's the demo?* → **A Gradio web app** (Hugging Face Space) that runs the exact trained pipeline on an

uploaded photo and shows AQI + interval + category + an answer/abstain badge.

Practical & scope

- *What dataset, and how big?* → **PM25Vision:** 11,096 photos across 3,259 monitoring stations after

dedup; imagery from Mapillary, labels from WAQI.

- *How big is the model and how long to train?* → **EfficientNet-B0, $\sim 5.3\text{M}$ parameters**; a single model

trains in ~ 30 minutes on a free T4 GPU; cross-validation is 5 such runs.

- *Why a 0.65 / 0.15 / 0.20 split?* → ****Enough to train, a dedicated calibration set for the conformal**

guarantee, and a 20% held-out test evaluated once.**

- *What exactly is the calibration set for?* → ****Two honesty steps that must never see training or test**

data:** computing the conformal widening Q , and fitting the isotonic map. Selecting on it also drives early stopping.

- *Why the 5th/95th percentiles, not 10th/90th?* → **5/95 gives a nominal 90% central interval**, which

matches our target coverage; conformal then adjusts it to the exact guarantee.

- *Could you just use more data / a bigger model?* → **More honest data would help most** (the bottleneck

is leakage-safe training images, not compute); a bigger backbone is a config change we can add as a new row, but the paper found B0 best at this scale.

- *How is this different from ordinary dehazing?* → **We don't remove haze; we read it.** The Dark Channel

Prior gives us a haze *measurement* (transmission) that we feed the model as a feature.

- *Is the abstention just a confidence threshold?* → **No — it's two gates:** an out-of-distribution gate

first (station-invariant "too dark / too flat" checks, for unusable photos), then a calibrated uncertainty gate (for valid but hard photos).

- *What would make this production-ready?* → **A second dataset for external validity, group-conformal for

per-region coverage, the C3 run, and a larger honest training set;** the architecture and honesty machinery are already in place.

- *What is the single biggest limitation?* → **The task is intrinsically hard** (one instant vs a

daily-average label across new places), so honest accuracy is modest; we are transparent about that and bound it with the C2 ceiling.

10.3 Traps & honest caveats (own them before you're asked)

1. R^2 depends on the test set's variance — always cite MAE + SD next to it.
2. "R²" is ambiguous — we report the strict coefficient of determination **and** squared-Pearson.
3. The -0.86 clean R^2 is **variance-amplified**; lead with the MAE gap (25.4 vs 30.4).
4. 0.55 is **not** unreachable honestly (a fold hit 0.574) — the claim is unreliability + leakage, not

impossibility.

1. The CV CI is mildly optimistic (correlated folds); the rigorous fix is repeated CV + Nadeau–Bengio.
2. The ceiling is an *upper bound* (~0.98), split-specific, with the top ~13% extrapolated conservatively.
3. Coverage $0.87 < 0.90$ on one split is expected drift (broken exchangeability on unseen stations), not a

bug; CV coverage 0.895 confirms the method.

1. The point head is **MSE + standardise + isotonic** — not the median, and not the old Huber approach.
 2. We never tune on test — every choice is made on calibration/CV, and test is evaluated once.
 3. We don't accuse anyone of cheating; we measure a mechanism the benchmark's unspecified split admits.
-

11. Who did what — the 4-person work split

(Full detail, with each member's files and likely questions, is in [docs/TEAM_CONTRIBUTIONS.md](#).)

Member	Owns	Presents
Keyaan	Data load/clean/audit, leakage-safe splits, the leakage experiment, integration + results ledger	The central leakage finding + the honest-evaluation thesis
Anup	Physics features (DCP + inverted saturation), EfficientNet-B0 + quantile/point heads, losses, training	How the model works
Lamiya	Conformal calibration, isotonic recalibration, metrics, cross-validation, the correctness audits	How we know the numbers are honest and the intervals hold
Asif	C2 error ceiling, C3 abstention, the inference pipeline + Gradio demo (HF Space)	The ceiling, abstention, and the live demo

Everyone should know the **five-minute story** and the **top-level thesis**, so any member can field a cross-cutting question.

12. Glossary (say-it-in-one-breath, then deeper)

- **AQI** — one-breath: the pollution number apps show (1–530). Deeper: a piecewise-linear index of

pollutant concentration; ours is the daily-average PM2.5 AQI.

- **Model / parameters / feature / label** — a tunable function / its adjustable numbers / the input it

reads / the correct answer.

- **Training / gradient descent / epoch** — tuning parameters to shrink the loss / the small-step

optimisation that does it / one pass over the data.

- **Generalisation / overfitting** — doing well on unseen inputs / memorising the training set.
- **CNN / filter / feature map / pooling** — an image network / a small sliding detector / its output map /

a shrink-and-summarise step.

- **Backbone / pretrained / transfer learning** — a reusable CNN body / already trained on ImageNet /

adapting those weights to our task.

- **MAE / RMSE** — average absolute miss / root-mean-square miss (punishes big misses more).
- **R² (coefficient of determination)** — fraction of variance explained vs guessing the mean; can be

negative; depends on test-set spread.

- **r²(Pearson)** — squared correlation; ranking only; always $\geq R^2$.
- **Spearman** — rank correlation; are photos ordered correctly?
- **Leakage** — test info reaching training, so the score measures memorisation.
- **Station-grouped split / GroupKFold** — all of one station's photos in one split / the cross-validation

version (rotate which fifth of the *stations* is tested).

- **Δ_{leak}** — how much a leaky split inflates the score over the honest one.
- **Dark Channel Prior / transmission** — the haze-estimation trick / low = dense haze.
- **Inverted saturation** — high where colour is washed out (hazy/sky).
- **Quantile / pinball loss** — a "fraction-below" value / the asymmetric loss that teaches a percentile.
- **Softplus / monotone heads** — a smooth positive function / heads whose outputs can't cross by

construction.

- **Conformal prediction (CQR) / exchangeability** — the procedure that guarantees ~90% coverage for any

model / the assumption (cal and test drawn alike) it needs.

- **Isotonic recalibration** — a monotone prediction→truth map that removes systematic bias.
- **Coverage / width** — how often the truth is inside the range / how wide the range is.
- **R²_max (error ceiling) / Var(ϵ) / Var(y)** — the best R² achievable given daily-average labels (an

upper bound, ~ 0.98) / within-day label noise / label spread.

- **Abstention / OOD / Mahalanobis distance** — refusing to answer / an input unlike the training data /

a shape-aware distance in feature space used to detect OOD.

- **AURC / AUROC / FPR@95** — area under the risk–coverage curve (abstention quality) / OOD-ranking

quality / false-refusal rate at 95% true-positive rate.

13. Design decisions and the alternatives we rejected

A professor's favourite move is "why did you do X and not Y ?". Here is every major choice, what we rejected, and the reason — so you can answer instantly. The meta-principle: ****every choice defends honesty and generalisation****, sometimes at the cost of a higher-but-fake number.

Decision	We chose	We rejected	Why
Evaluation split	station-grouped	random split	random puts same-station photos on both sides → leakage → inflated score
Point objective	MSE on a standardized target (mean)	median / Huber	the median under-shoots on right-skewed labels and depresses R^2
Interval method	conformal (CQR)	Gaussian / Bayesian intervals	CQR <i>guarantees</i> ~90% coverage for any model, distribution-free
Quantile ordering	cumulative softplus (by construction)	penalty term or post-hoc sort	guaranteed non-crossing with no extra loss and no hacks
Backbone	EfficientNet-B0	ResNet50 / ViT / bigger nets	the source paper found B0 best on ~7k images; extra capacity overfits
Input	5-channel (RGB + 2 physics maps)	RGB only	physics generalises across cities; RGB-only scored lower and leaks scenery
Augmentation	geometric only	photometric (brightness/contrast)	photometric jitter corrupts the exact haze signal we rely on
Quantile target space	log(AQI)	raw AQI	log stabilises the skewed target so a few extremes don't dominate
Robustness check	GroupKFold by station	plain k-fold	plain k-fold re-leaks same-station photos
OOD score	handcrafted station-invariant checks (too dark / too flat)	Mahalanobis on features; interval width	Mahalanobis over-refuses unseen-station photos (station shift); interval width is invalid off-distribution (conformal void there)
De-biasing	isotonic on calibration	none / parametric fit	monotone (can't reorder), removes tail bias, low-risk, no retraining
Ceiling noise estimate	hourly InstantCast, level-reweighted	unweighted average / NowCast	matches how the labels are built and our dataset's pollution mix

Decision	We chose	We rejected	Why
Early-stopping metric	calibration point-MAE	training loss / last epoch	we select on the metric we care about, on held-out data, and keep the best checkpoint

Key idea. If a professor asks "why not the higher-scoring option?", the answer is almost always:
"because that option leaks or over-fits, and our goal is an honest number, not a fake one."

14. Limitations and honest future work

Stating limitations *before* you're asked is the strongest defensive move — it shows you understand the work's edges. Ours:

- **Honest accuracy is modest ($R^2 \approx 0.385$).** Reading instantaneous pollution from a *single* photo of a

new location is genuinely hard; the honest number is field-normal, not a failure — but it is not high, and we say so. More photos per station, multi-photo aggregation, or temporal context would help.

- **The C2 ceiling's top ~13% is extrapolated.** The OpenAQ reference still under-covers the very-high-AQI

bands (200+); we impute them conservatively (the ceiling barely moves), but a targeted high-pollution fetch would let us measure them directly and tighten ~0.98.

- **The cross-validation CI is mildly optimistic.** Five folds from one GroupKFold partition are

correlated; a repeated grouped CV with the Nadeau–Bengio correction would give a fully rigorous interval. The point estimate (0.385) and the spread (0.15–0.57) are unaffected.

- **Coverage drifts under distribution shift.** On unseen stations, exchangeability is mildly broken, so

single-split coverage is 0.87 rather than 0.90. Group-balanced or Mondrian conformal could recover the target per subgroup.

- **The C3 abstention numbers are pending one run.** The method is built and locally tested; the AURC /

AUROC / FPR95 values land after **08_abstention.ipynb** runs on the three external OOD sets.

- **We evaluate one dataset.** PM25Vision is large and global, but a second dataset would strengthen the

external validity of both the honest number and the leakage effect.

Future work (all low-risk extensions): a larger backbone reported as a *new row* beside B0 (both on record); additional physics channels; multi-photo or temporal aggregation to beat the single-instant ceiling; group-conformal for per-region coverage; and a repeated-seed CV for the rigorous CI.

Appendix A — the number sheet (every headline figure)

- **Data:** 11,096 photos / 3,259 stations (median 2, max 91); 123 duplicate image IDs removed;

falsification Pearson $r \approx -0.14$.

- **Honest single split (station_grouped):** R^2 0.220 · MAE 54.2 · RMSE 95.5 · Spearman 0.649 · coverage

0.872 · SD(y) 108.1 · n_{train} 7,217 · n_{test} 2,216.

- **Leakage gradient (R^2):** random 0.759 · temporal 0.380 · shipped 0.378 · station_grouped 0.220 ·

geographic -0.079 . $\Delta_{\text{leak}}(\text{MAE}) = 27.8$ AQI; $\Delta_{\text{leak}}(R^2) = 0.539$ (vs 0.220) / 0.374 (vs 0.385).

- **Causal test:** contaminated R^2 0.762 (n 1,757, MAE 25.4) vs clean R^2 -0.86 (n 463, MAE 30.4).
- **Cross-validation:** R^2 0.385 (SD 0.15; 95% CI [0.196, 0.574]) · MAE 46.3 · Spearman 0.679 · coverage

0.895; folds {0.405, 0.574, 0.365, 0.429, 0.152}.

- **C2 ceiling:** $R^2_{\text{max}} \approx 0.98$ (0.982 optimistic / 0.979 conservative; 0.980 under 2024 breakpoints);

$\text{Var}(\epsilon)$ 208.6–247.9; $\text{Var}(y)$ 11,708; interval-width floor ≈ 35 AQI; 32 stations / 6 countries / 1,241 station-days.

- **Benchmark:** EfficientNet-B0 R^2 0.550 (MAE 36.6, RMSE 54.6). Closest honest analogue Mondal 2024 $\approx$

0.39.

- **C3:** built; risk–coverage/AURC + OOD AUROC/FPR95 land after the **08_abstention.ipynb** run.

Appendix B — the equation sheet

- MAE, RMSE, R^2 , and the identity $R^2 \approx 1 - (\text{RMSE}/\text{SD})^2$ — Section 2.7.
- Hazy-image model $I = J \cdot t + A(1-t)$ and the DCP transmission — Section 5.1.
- Monotone quantiles via cumulative softplus — Section 5.3.
- Standardise/de-standardise the point head — Section 5.4.
- Pinball loss L_τ — Section 5.5.
- Conformal score E_i and widening Q — Section 6.1.
- Error ceiling $R^2_{\text{max}} = 1 - \text{Var}(\epsilon)/\text{Var}(y)$ and $\text{Var}(\epsilon) = \sum p(m)v(m)$ — Section 8.1.
- Mahalanobis distance D_M — Section 8.2.